

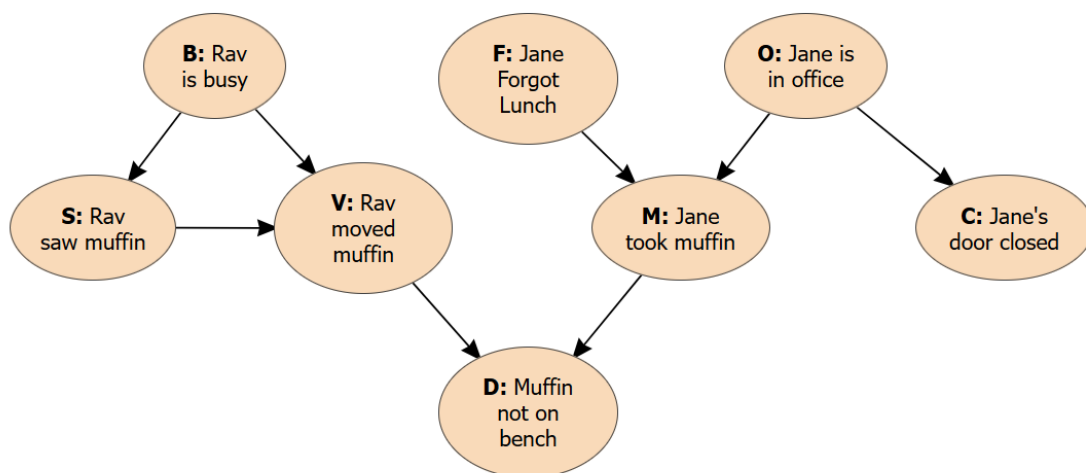
Monash University, Faculty of Information Technology

FIT3080

Lab: Bayesian Networks – d-separation, Inference & Decision Networks

Exercise 1: d-separation (& d-connection)

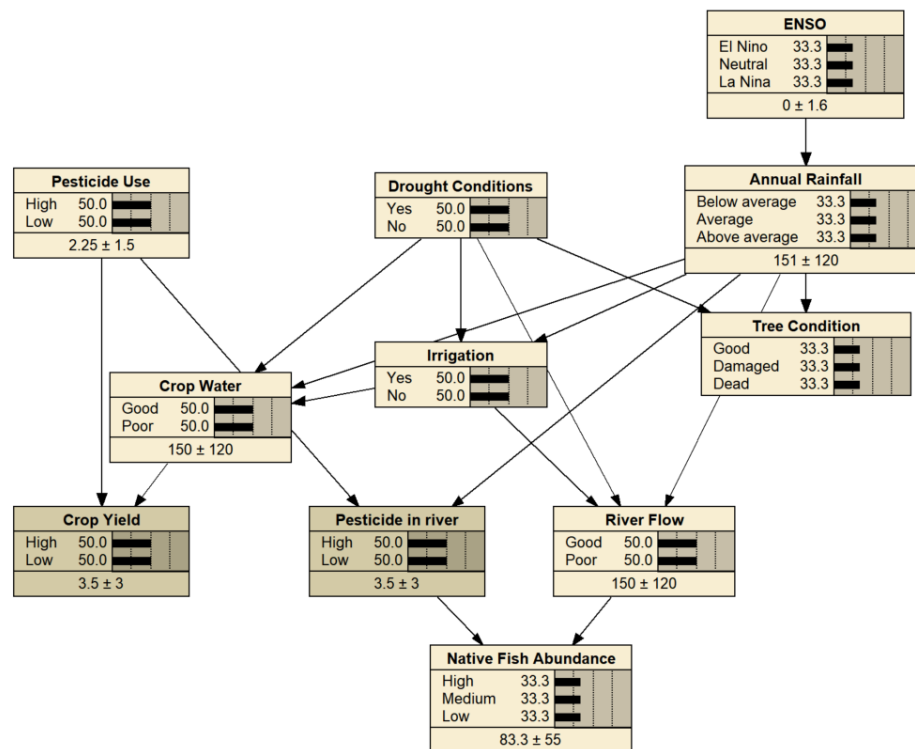
Scenario: You've brought a muffin to work and accidentally left it out on the bench before heading out for an appointment. Jane has a habit of taking food that's out if she's in the office and has forgotten her lunch. Rav will move the muffin somewhere to keep it safe for you, but only if he sees it and he's not busy. While waiting at your appointment, you sketch the following model:



- What nodes are d-separated from D?
- What nodes are d-separated from D given you observe M?
- Is B d-separated or d-connected to D when you observe S and M? Why?
- What nodes are d-connected to C?
- What nodes would be d-connected to C if you observed M?
- Are F and O d-connected if you observe i) D, ii) V, iii) D and V?
- (Optional) Discuss the answers in (f), in the context of the problem. For example, why does observing whether or not there's a muffin on the bench (D) change the probabilistic relationship between Jane forgetting her lunch (F) and Jane being in the office (O)?

Exercise 2: Learning, reasoning and evaluation

An extended version of the native fish model is shown below:



This model contains some nodes where numeric intervals have been provided for the states of the node. In addition to the usual probabilities, Netica will display an expected value for these nodes (e.g. 3.5 ± 3 for Crop Yield) at the bottom of the node.

You've been given three datasets that have been prepared to work with this BN. Both the unparameterised model (**nfv3_unparameterised.dne**) and the datasets (**nfv3_complete.csv**, **nfv3_missing.csv** and **nfv3_testing.csv**) can be downloaded via Moodle, in the applied class section for Week 7. (Look for "Week 7 Applied Class: Materials for Q2 - BNs and datasets".)

- a) Train the given model on **nfv3_complete.csv** using count learning, by using Netica's count learning function (menu: Cases -> Learn -> Incorp Case File. Leave the 'Enter degree' prompt at 1)
 - i. How does native fish abundance – both the *probability distribution and the expected value* – change when you observe low pesticide use?
 - ii. How does it change when you also know that tree condition is not good? (Note: shift-click to set negative evidence in Netica)
- b) Clear all evidence (menu: Cases -> Remove findings), and run a sensitivity to findings (menu: Network -> Sensitivity to findings) on the native fish abundance node.
 - i. What 3 nodes have the strongest influence on native fish abundance? (Not including native fish abundance itself)
 - ii. How does this change if river flow is good? Why might explain the change?

- c) Clear all evidence, and evaluate the performance of the model in predicting native fish abundance (menu: Cases -> Test With Cases) using **nfv3_complete.csv**.
 - i. What is the error rate, confusion matrix and log score?
 - ii. Why might using **nfv3_complete.csv** not be the best way to test the model?
 - iii. Now test the performance with **nfv3_testing.csv**. (50% of the values in this file are missing. You can inspect the file in a spreadsheet tool to see them.) What is the new error rate, confusion matrix and log score? How does it compare to the test that used **nfv3_complete.csv**?
- d) Remove all the CPTs (menu: Table -> Remove) and use count learning again to train the model with **nfv3_missing.csv**. (A high 80% of the values in this dataset are missing.) Test the newly trained model's ability to predict native fish abundance with **nfv3_testing.csv**.
 - i. What is the error rate, confusion matrix and log score?
 - ii. What do the results in the confusion matrix suggest?
 - iii. How does this compare to the results from (c) (iii)? How would you explain the difference in the results?
- e) Repeat (d), but this time use expectation-maximisation (menu: Cases -> Learn -> Learn using EM; this may take 1-2 minutes). How do the results compare?

(Note that the Netica demos in the recorded lecture may help you with Q2.)

Exercise 3: Decision Networks, Netica¹

Clare's football team, Melbourne, is going to play her friend John's team, Carlton. If the weather is wet, Melbourne has a 60% chance of winning, but if the weather is dry, Melbourne's chances of winning are only 25%. At this time of the year, there is a 30% chance of rain on any day.

John offers Clare a friendly bet: whoever's team loses will buy the wine next time they go out for dinner (at a cost of \$15).

To decide whether to accept the bet, Clare will assess her team's chances of winning. She also knows that she will be happy if her team wins and unhappy if her team loses, regardless of the bet.

1. Using the variables *Weather* and *Result* (for Melbourne) and the decision node *Accept* bet, **build a BDN for this scenario using Netica software**. Add utilities according to the narrative.
2. Determine whether Clare should make the bet for each weather condition.

¹From Korb & Nicholson, 2010 2nd Edition.

Exercise 4: Decision Networks, Netica²

Robert is trying to decide whether to study hard for the AI exam. He wants to get a High Distinction (HD) for the unit (he is not interested in any other marks), but he knows that his mark will depend not only on how hard he studies but also on how hard the exam is, and if he did well in the Theory of Computation unit (which indicates how well prepared he is for the current unit). There is a 50-50 chance that the AI exam will be hard, and 10% of all students receive an HD in Theory of Computation.

In addition, Robert would prefer to get a HD without studying hard.

1. State which nodes are required in a Bayesian Decision Network, what are their types, and what values they should take.
2. What is Robert's utility function? What chance nodes, if any, will it depend on? Does it depend on the decision node?
3. **Build a Decision network to model this problem using the Netica Software:** insert arcs that connect the chance, decision and utility nodes. Quantify the relationships in the network by populating the CPTs of chance nodes and the utility table for the utility node (use your own intuitions to come up with suitable values). Do you need an information link between the evidence node and the decision node?
4. Once you have built your model, use Netica to show the beliefs for the chance nodes and the expected decisions for the utilities before any evidence is added. Add the evidence regarding Robert's mark in Theory of Computation, and use Netica to see how the beliefs and the decision change, if at all. Explain your results.

² Based on Korb & Nicholson, 2010 2nd Edition, 4.10, Exercise 3.